

## 1. DESCRIPTION

MT5706 is a System on Chip (SoC) for magnetic induction based wireless power receiver.

It is capable of wireless charging for up to 24W(12V/2A) and can be configured as a wireless charging transmitter to charge other receivers. It is fully compliant with WPC Qi Specification (Version 1.3) of BPP (Baseline Power Profile) and EPP (Extended Power Profile) and also supports various proprietary fast charging protocols used by major smart phone OEM's.

MT5706 has a very high overall AC to DC conversion efficiency (up to 97%), thanks to the optimized and adaptive full synchronous rectifier control, very small  $R_{DS(on)}$  of power MOSFET, and extremely low bias current are achieved.

The AC input can be converted to fully programmable DC output voltage due to the magnetic induction charging technology.

Designed with ARM Cortex M0 processor, integrated with optimized and adaptive full synchronous rectifier control and special LDO, the chip achieves high efficiency, ultra-low bias current and very small power MOSFET  $R_{DS(on)}$ .

MT5706 is embedded with various protection features, such as FOD, over-voltage, over-current and over-temperature protection which guarantee the system reliability.

## 2. APPLICATIONS

- Smartphones and wearable devices with high integration and small form factor
- Rx function for power banks where they can be wirelessly charged
- Other wireless power applications

## 3. FEATURES

- Power delivery: up to 24W(12V/2A)
- Power delivery in Tx mode: up to 5W
- Fully programmable output voltage and current limit
- Embedded with ARM Cortex M0 processor with 8KB SRAM and 32KB MTP
- Supports half-wave mode (default) and full-wave mode
- 85KHz~2MHz wide wireless charging frequency
- Supports 1.8V I/O and 1.2V I/O
- Embedded with various protection features:
  - over-voltage protection (OVP)
  - over-current protection (OCP)
  - short-circuit protection (SCP)
  - over-temperature protection (OTP)
  - foreign object detection (FOD)
- Innovative output LDO with output clamping and fast response to line and load transient
- Qi 1.3 compliant and proprietary communication protocols support with hardware ASK modulation
- Independent I<sup>2</sup>C slave interface with additional GPIO's
- Halogen free and RoHS compliant
- 2.10mm x 3.30mm (5 x 8 ball array) 40-WLCSP

#### 4. TYPICAL APPLICATION CIRCUIT

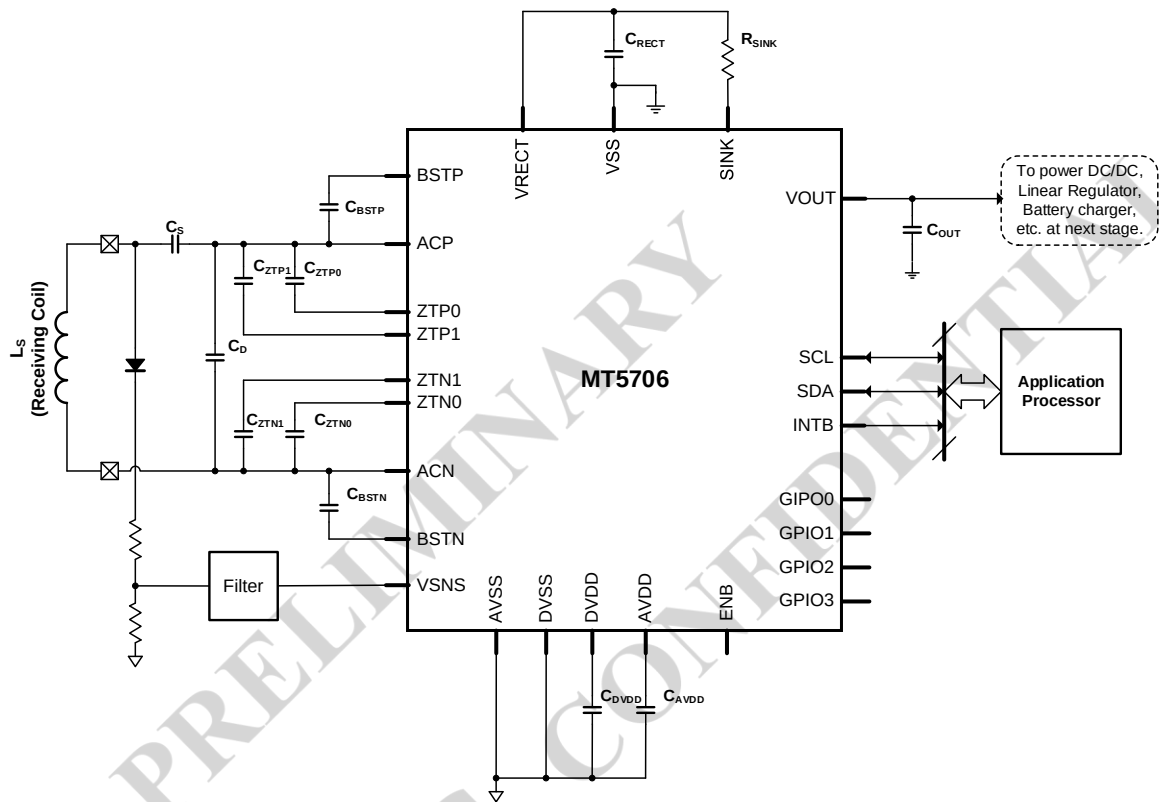


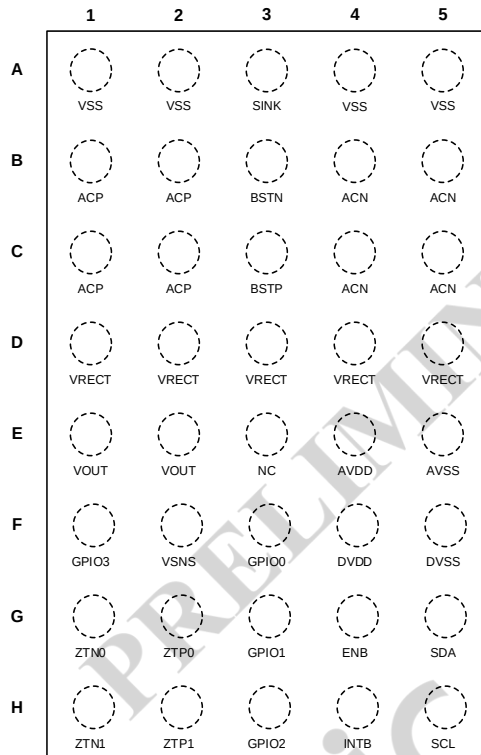
Figure 1 Typical Application Circuit

## Content

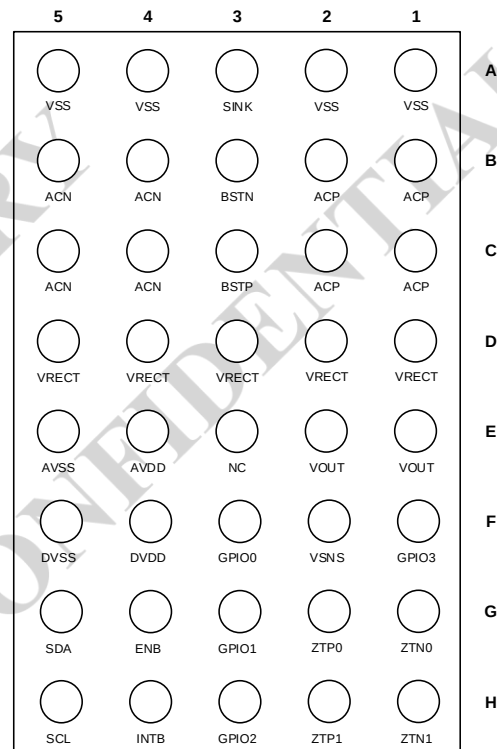
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## 5. PIN CONFIGURATIONS AND FUNCTIONS

### 5.1. Pin Configuration



**Figure 2 Top View**



**Figure 3 Bottom View**

## 5.2. Pin Functions

**Table 1 Pin Functions**

| Name  | Location           | Description                                                                                                                      |
|-------|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| ACN   | B4, B5,C4,C5       | AC negative input.                                                                                                               |
| ACP   | B1, B2,C1,C2       | AC positive input. ACP is internally connected to GND in the half-wave mode.                                                     |
| VSS   | A1, A2,A4,A5       | Power ground.                                                                                                                    |
| AVSS  | E5                 | Analog ground                                                                                                                    |
| DVSS  | F5                 | Digital ground                                                                                                                   |
| VRECT | D1, D2, D3, D4, D5 | Output of synchronous rectifier. Recommend to connect three 10 $\mu$ F capacitors between this pin and GND.                      |
| VOUT  | E1, E2             | Output of power LDO. Recommend to connect two 10 $\mu$ F capacitors between this pin and GND.                                    |
| SINK  | A3                 | Providing sinking current. Recommend to connect a 50 $\Omega$ resistor between this pin and VRECT.                               |
| BSTP  | C3                 | Boost capacitor for internal driver for bridge rectifier at ACP. Recommend to connect a 47nF capacitor between this pin and ACP. |
| BSTN  | B3                 | Boost capacitor for internal driver for bridge rectifier at ACN. Recommend to connect a 47nF capacitor between this pin and ACN. |
| ZTP0  | G2                 | ASK modulation FET at ACP. Recommend to connect a 10nF capacitor between this pin and ACP.                                       |
| ZTN0  | G1                 | ASK modulation FET at ACN. Recommend to connect a 10nF capacitor between this pin and ACN.                                       |
| ZTP1  | H2                 | ASK modulation FET at ACP. Recommend to connect a 22nF capacitor between this pin and ACP.                                       |
| ZTN1  | H1                 | ASK modulation FET at ACN. Recommend to connect a 22nF capacitor between this pin and ACN.                                       |
| EN_B  | G4                 | Disable pin. Connect this pin to logic HIGH can disable the whole chip, while to logic LOW or floating can enable the chip.      |
| AVDD  | E4                 | Internal 5V power supply. Recommend to connect a 4.7 $\mu$ F capacitor between this pin and GND.                                 |
| DVDD  | F4                 | Internal 1.5V power supply. Recommend to connect a 1 $\mu$ F capacitor between this pin and GND.                                 |
| SDA   | G5                 | I <sup>2</sup> C slave SDA.                                                                                                      |
| SCL   | H5                 | I <sup>2</sup> C slave SCL.                                                                                                      |
| INTB  | H4                 | I/O & Interrupt. This is the interrupt from wireless power to AP.                                                                |
| GPIO0 | F3                 | General purpose I/O, refer to section <a href="#">7.2.3</a> for more details.                                                    |
| GPIO1 | G3                 |                                                                                                                                  |
| GPIO2 | H3                 |                                                                                                                                  |
| GPIO3 | F1                 |                                                                                                                                  |

| Name | Location | Description                                                               |
|------|----------|---------------------------------------------------------------------------|
| VSNS | F2       | Coil's voltage sense for demodulation as a Tx, can be configured as GPIO. |
| NC   | E3       | Floating pin.                                                             |

## 6. SPECIFICATIONS

### 6.1. Absolute Maximum Ratings

Table 2 Absolute Maximum Ratings

| Pin Name                                        | Rating         |
|-------------------------------------------------|----------------|
| ACN, ACP<br>ZTP0, ZTN0, ZTP1, ZTN1              | -0.3V to 20V   |
| BSTP with respect to ACP                        | -0.3V to 5.5V  |
| BSTN with respect to ACN                        | -0.3V to 5.5V  |
| VRECT                                           | -0.3V to 20V   |
| VOUT                                            | -0.3V to 20V   |
| SINK                                            | -0.3V to 20V   |
| AVDD                                            | -0.3V to 5.5V  |
| DVDD                                            | -0.3V to 1.65V |
| SCL, SDA, INTB, VSNS                            | -0.3V to 5.5V  |
| GPIO0~3                                         | -0.3V to 5.5V  |
| ENB                                             | -0.3V to 5.5V  |
| Storage Temperature (T <sub>STG</sub> )         | -55°C to 150°C |
| Maximum Soldering Temperature (Reflow, Pb-Free) | 260°C          |

**Note:**

- Stresses greater than those listed as Absolute Maximum Ratings could cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated in the operational sections of this specification. Exposure to absolute maximum rating conditions for extended periods might affect reliability.
- All voltages are referred to ground unless otherwise stated.

## 6.2. ESD Ratings

Table 3 ESD Ratings

| Test model | Pins     | Rating |
|------------|----------|--------|
| HBM        | All pins | 2000V  |
| CDM        | All pins | 1000V  |
| LU         | All pins | 250mA  |

## 6.3. Recommended Operating Conditions

Table 4 Recommended Operating Conditions

| Parameter                           | Rating       |
|-------------------------------------|--------------|
| Operating Temperature (Environment) | -40°C ~ 85°C |
| Operating Current ( $I_{OUT}$ )     | 0A ~ 1.5A    |
| Operating Voltage ( $V_{OUT}$ )     | 4.5V ~ 10V   |

## 6.4. Package Thermal Information

Table 5 Package Thermal Information

| Parameter                               | Value  |
|-----------------------------------------|--------|
| Junction to ambient ( $R_{\theta JA}$ ) | 48°C/W |

## 6.5. Electrical Characteristics

Test conditions:  $V_{RECT}=5.5V$ ,  $T_A=25^{\circ}C$ , unless otherwise stated.

**Table 6 Electrical Characteristics**

| Symbol                         | Parameters                                                    | Conditions                             | Min.  | Typ.  | Max.  | Unit       |
|--------------------------------|---------------------------------------------------------------|----------------------------------------|-------|-------|-------|------------|
| <b>Startup</b>                 |                                                               |                                        |       |       |       |            |
| $V_{UVLO}$                     | Under voltage lockout threshold                               | $V_{RECT}$ rising from 0V              | 2.1   | 2.35  | 2.6   | V          |
| $V_{UVLO\_HYS}$                | Hysteresis of under voltage lockout                           | $V_{RECT}$ falling                     |       | 90    |       | mV         |
| <b>Supply Current</b>          |                                                               |                                        |       |       |       |            |
| $I_{Q\_OP}$                    | Quiescent current in normal operation mode                    | $V_{RECT}=6V$                          |       |       | 10    | mA         |
| $I_{SLEEP}$                    | Supply current from $V_{RECT}$ pin when chip is in sleep mode | $V_{RECT}=6V$                          |       |       | 0.5   | mA         |
| <b>ENB</b>                     |                                                               |                                        |       |       |       |            |
| $I_{ENB}$                      | Supply current from $V_{OUT}$ when chip is disabled           | $V_{OUT}=5V$                           |       |       | 320   | $\mu A$    |
| <b>Rectifier Bridge</b>        |                                                               |                                        |       |       |       |            |
| $R_{BRG}$                      | $R_{DS(ON)}$ of power MOSFETs in the rectifier bridge         |                                        |       | 45    | 60    | m $\Omega$ |
| <b>Over-voltage Protection</b> |                                                               |                                        |       |       |       |            |
| $V_{OVP1}$                     | OVP threshold for $V_{RECT}$ , trigger sink                   | programmable, ranges from 11V to 18.5V |       | 16    |       | V          |
| $V_{OVP2}$                     | OVP threshold for $V_{RECT}$ , short low-side MOS             | programmable, ranges from 11V to 18.5V |       | 18    |       | V          |
| $V_{LSB\_OVP}^{①}$             | Least significant bit for OVP threshold programming           |                                        |       | 500   |       | mV         |
| $T_{OVP\_RES}$                 | OVP response time <sup>①</sup>                                |                                        |       |       | 300   | ns         |
| <b>Power LDO</b>               |                                                               |                                        |       |       |       |            |
| $V_{OUT\_5V}$                  | Output voltage when programming to 5V                         | $V_{RECT}=5.5V$ , $I_{OUT}=10mA$       | 4.95  | 5.00  | 5.05  | V          |
| $V_{OUT\_9V}$                  | Output voltage when programming to 9V                         | $V_{RECT}=10V$ , $I_{OUT}=10mA$        | 8.95  | 9.00  | 9.05  | V          |
| $V_{OUT\_12V}$                 | Output voltage when programming to 12V                        | $V_{RECT}=13V$ , $I_{OUT}=10mA$        | 11.95 | 12.00 | 12.05 | V          |
| $V_{LSB\_VOUT}^{①}$            | Least significant bit for output voltage programming          |                                        |       | 25    |       | mV         |
| $V_{PR\_VOUT}^{①}$             | Programming range of output voltage                           |                                        | 3     |       | 12    | V          |

| Symbol                             | Parameters                                            | Conditions                                                                  | Min. | Typ. | Max. | Unit |
|------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------|------|------|------|------|
| R <sub>DS(ON)</sub>                | R <sub>DS(ON)</sub> of power MOSFETs in the power LDO |                                                                             |      | 70   | 85   | mΩ   |
| Modulation FET                     |                                                       |                                                                             |      |      |      |      |
| R <sub>DS(ON)_MOD</sub>            |                                                       |                                                                             |      | 0.95 | 1.6  | Ω    |
| ADC                                |                                                       |                                                                             |      |      |      |      |
| N <sup>①</sup>                     | Resolution                                            |                                                                             |      | 12   |      | Bit  |
| f <sub>SMP</sub>                   | Sampling rate                                         |                                                                             |      | 67   |      | kS/s |
| N <sub>CHN</sub>                   | Number of channels                                    |                                                                             |      | 8    |      | --   |
| ACC <sub>VRECT</sub>               | Sampling accuracy for V <sub>RECT</sub>               | V <sub>RECT</sub> =9V                                                       | -1   |      | 1    | %    |
| ACC <sub>VOUT</sub>                | Sampling accuracy for V <sub>OUT</sub>                | V <sub>OUT</sub> =9V                                                        | -1   |      | 1    | %    |
| ACC <sub>IOUT</sub>                | Sampling accuracy for I <sub>OUT</sub>                | I <sub>OUT</sub> <500mA                                                     | -10  |      | 10   | mA   |
|                                    |                                                       | 500mA≤I <sub>OUT</sub> <1500mA                                              | -2   |      | 2    | %    |
| Miscellaneous                      |                                                       |                                                                             |      |      |      |      |
| V <sub>AVDD</sub>                  | Output voltage of AVDD                                | V <sub>RECT</sub> =6V, AVDD load=10mA                                       |      | 5.0  |      | V    |
| V <sub>DVDD</sub>                  | Output voltage of DVDD                                | V <sub>RECT</sub> =5V, DVDD load=10mA                                       |      | 1.5  |      | V    |
| f <sub>OSC</sub>                   | Internal oscillation frequency                        |                                                                             | 78.5 | 80   | 81.5 | MHz  |
| Digital I/O Pins <sup>②</sup>      |                                                       |                                                                             |      |      |      |      |
| V <sub>IH</sub>                    | Input voltage threshold for logic High                | 1.8V Level                                                                  | 1.1  |      |      | V    |
|                                    |                                                       | 1.2V Level                                                                  | 0.7  |      |      |      |
| V <sub>IL</sub>                    | Input voltage threshold for logic Low                 | 1.8V Level                                                                  |      |      | 0.7  | V    |
|                                    |                                                       | 1.2V Level                                                                  |      |      | 0.46 |      |
| V <sub>OH</sub>                    | High level output voltage <sup>③</sup> of GPIO0,1,2   | I <sub>SINK</sub> =0.6mA                                                    | 1.2  |      |      | V    |
|                                    | High level output voltage <sup>③</sup> of GPIO3       | I <sub>SINK</sub> =2mA                                                      | 1.2  |      |      | V    |
| V <sub>OL</sub>                    | Low level output voltage                              | I <sub>SOURCE</sub> =2mA                                                    |      |      | 50   | mV   |
| I <sub>GPIO_LKG</sub> <sup>③</sup> | Leakage current of GPIO0,1,2                          | V <sub>GPIOX</sub> =1.8V                                                    |      |      | 1    | μA   |
|                                    | Leakage current of GPIO3                              | V <sub>GPIO3</sub> =1.8V                                                    |      |      | 25   | μA   |
| Reverse Charging Mode              |                                                       |                                                                             |      |      |      |      |
| V <sub>IN_MAX</sub>                | Maximum input voltage                                 | Powered by VOUT.<br>The parameter is programmable, ranges from 0V to 12.5V. |      | 9.5  |      | V    |
| V <sub>IN_UVLO</sub>               | Under voltage lockout threshold of input voltage      | Powered by VOUT.                                                            | 3.1  |      |      | V    |
| I <sub>OCP_PEAK</sub>              | Peak of over current protection threshold             |                                                                             |      | 4    |      | A    |

| Symbol               | Parameters                                   | Conditions | Min. | Typ. | Max. | Unit |
|----------------------|----------------------------------------------|------------|------|------|------|------|
| I <sub>OCP_AVG</sub> | Average of over current protection threshold |            |      | 1.5  |      | A    |

**Note:**

- ①: Guaranteed by design.
- ②: All I/Os can be configured to open-drain type.
- ③: Only for push/pull type I/O.

## 7. DETAILED DESCRIPTIONS

### 7.1. Functional Block Diagram

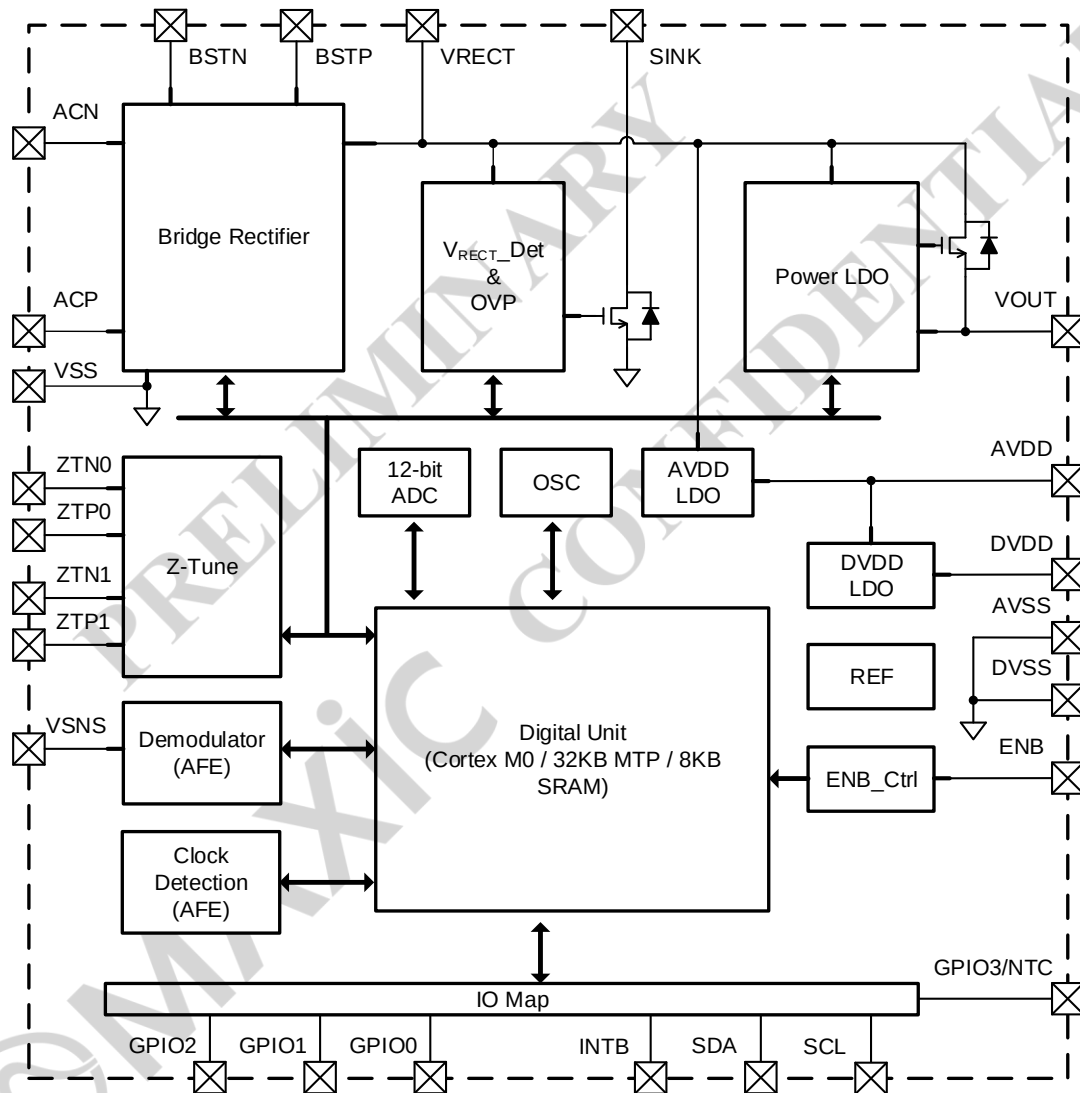


Figure 4 Functional Block Diagram

## **7.2. Theory of Operation**

MT5706 is a System on Chip (SoC) for wireless power receiver. It only needs several passive components, like power receiving coil, resonant tank capacitor, decoupling capacitor and pull-up/down resistors to build a complete wireless power receiver system. When coupled with a wireless power transmitter, this system can provide all the functions for wireless power transfer, including power receiving and rectification, output regulation, communication for power control and data exchange, and abnormal condition (FOD, over voltage, current, temperature, etc.) protection.

MT5706 is by default programmed to be fully compliant with WPC Qi Specification (Version 1.3) with support of both BPP (Baseline Power Profile) and EPP (Extended Power Profile). It can also be programmed to be compliant with major smart phone vendors' proprietary fast wireless charging protocols.

MT5706 is composed of several major functional blocks which together achieve the wireless power receiving function.

### **7.2.1. Rectifier Bridge**

MT5706 uses an integrated high efficiency full-wave synchronous rectifier bridge to convert AC signal from the resonant tank to DC signal on the VRECT pin. During startup, the rectifier bridge operates as a passive diode bridge, and goes into bridge rectifier mode as VRECT pin voltage exceeds  $V_{UVLO}$ . If VRECT pin voltage is too high, over-voltage protection is triggered.

Moreover, MT5706 is designed with an external wave-programmable rectifier bridge, whose operating mode can be configured to full-wave mode and half-wave mode. Once configured to half-wave mode, low power coil can be coupled.

### **7.2.2. LDO Regulator**

MT5706 integrates three LDOs: Power LDO, AVDD LDO and DVDD LDO.

- Power LDO creates power path from VRECT to VOUT\_RX. This block functions as a load switch (connecting and disconnecting the external load), output voltage and current regulation, and output clamping when fast load/line transient happens.
- AVDD LDO and DVDD LDO provide necessary regulated analog and digital power supplies separately from rectifier output for the operation of the chip.

### **7.2.3. GPIO**

General purpose I/O, which can be configured to GPOD mode. if operates as GPOD, the external pull-up resistor is necessary. MT5706 incorporates GPIO and GPIO\_BB, they have different structures and functions in IO or OD mode (refer to Figure 5). GPIO\_BB adopts back-to-back PMOS structure, which prevents leakage current when VDD50V is power down.



Figure 5 GPIO and GPIO\_BB structures

Table 7 I/O Pin Function and Configuration

| PIN      |       | Analog Function Map |     | Digital Function Map |       |        |       |
|----------|-------|---------------------|-----|----------------------|-------|--------|-------|
| Location | Name  | Type                | ADC | 0                    | 1     | 2      | 3     |
| F3       | GPIO0 | GPIO_BB             | ✓   | Hiz                  | input | output | inout |
| G3       | GPIO1 | GPIO_BB             | ✓   |                      |       |        |       |
| H3       | GPIO2 | GPIO_BB             | ✓   |                      |       |        |       |
| F1       | GPIO3 | GPIO                | ✓   |                      |       |        |       |

#### 7.2.4. ADC Converter

This block is one of the key blocks that converts various measured analog variables (voltage, current, temperature, external analog inputs, etc.) to digital domain such that the embedded microcontroller can use the information for follow-up actions.

#### 7.2.5. Digital Unit

This block contains all the digital circuits, including embedded microcontroller, volatile and non-volatile memories, I<sup>2</sup>C interfaces, peripherals, Direct Memory Access (DMA), internal buses, and other digital functional blocks. Digital unit is the brain of chip which dynamically configures the chip for different functions in different states, communicates with the external host, and performs necessary data processing for proper operation (such as FOD calculation, target  $V_{RECT}$  and  $V_{OUT}$  calculation, etc.).

## 7.2.6. I<sup>2</sup>C

### 7.2.6.1. I<sup>2</sup>C Interface

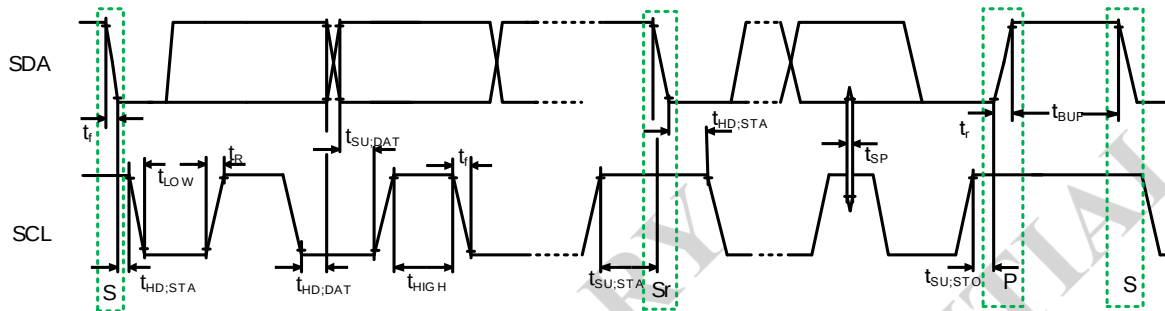


Figure 6 I<sup>2</sup>C Interface

### 7.2.6.2. I<sup>2</sup>C Timing Characteristics

All specifications are at  $V_{IO} = 1.8V$ ,  $T_{OPE} = 25^{\circ}C$ , unless otherwise noted.

Table 8 I<sup>2</sup>C Timing Characteristics

| Symbol       | Parameter (*)                                                                                  | Min | Max | Unit    |
|--------------|------------------------------------------------------------------------------------------------|-----|-----|---------|
| $f_{SCL}$    | SCL clock frequency                                                                            | 1   | 400 | kHz     |
| $t_{BUF}$    | Bus free time between STOP and the following START condition                                   | 1.3 |     | $\mu s$ |
| $t_{HD,STA}$ | Hold time (repeated) in START condition. After this period, the first clock pulse is generated | 0.6 |     | $\mu s$ |
| $t_{LOW}$    | Low period of the SCL clock                                                                    | 1.3 |     | $\mu s$ |
| $t_{HIGH}$   | High period of the SCL clock                                                                   | 0.6 |     | $\mu s$ |
| $t_{SU,STA}$ | Set-up time for a repeated START condition                                                     | 0.6 |     | $\mu s$ |
| $t_{SU,STO}$ | Set-up time for STOP condition                                                                 | 0.6 |     | $\mu s$ |
| $t_r$        | Rise time of both SDA and SCL signals                                                          |     | 300 | ns      |
| $t_f$        | Fall time of both SDA and SCL signals                                                          |     | 300 | ns      |
| $t_{HD,DAT}$ | Data hold time                                                                                 | 50  |     | ns      |
| $t_{SU,DAT}$ | Data setup time                                                                                | 100 |     | ns      |
| $t_{SP}$     | Pulse width of spikes which must be suppressed by the input filter                             | 0   | 50  | ns      |

**Note:**

(\*) Specified by design and characterization; not production tested.

### 7.2.6.3. ID and Address

I<sup>2</sup>C ID has 7 bits and identifies which I<sup>2</sup>C is valid. The last bit is the read/write bit and is appended by the host. The default I<sup>2</sup>C slave address is 0101011X, where X indicates read and write privileges, can be configured as:

- 0: write, equals to 0x56 in HEX.
- 1: read, equals to 0x57 in HEX.

**7.2.6.4. I<sup>2</sup>C Write/Read Protocol**

Write: 0x56 + ADDR[15:8] + ADDR[7:0] + DATA0 .... + DATAn

Read: 0x56 + ADDR[15:8] + ADDR[7:0] + 0x57 + DATA0 .... + DATAn

**7.2.7. Foreign Object Detection (FOD)**

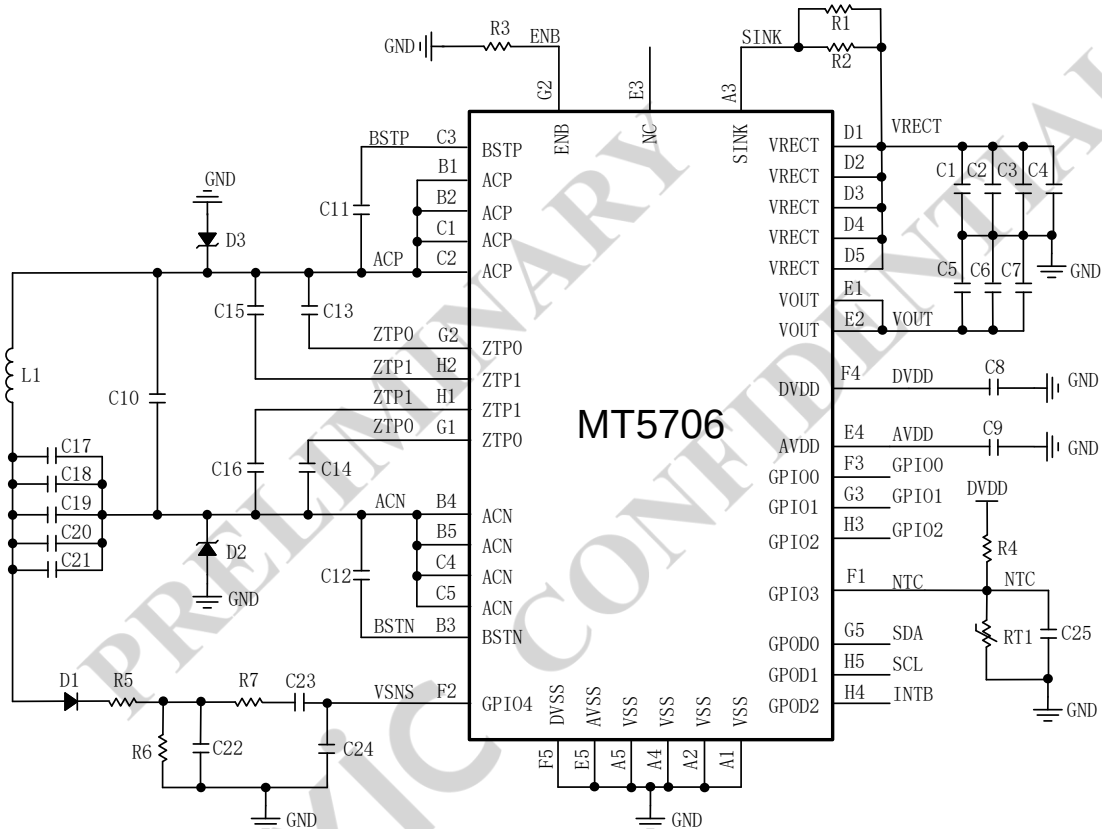
During power transfer phase, the receiver periodically communicates with the transmitter about the amount of power received, the transmitter compares this power with the power transmitted at the same time. By comparing the power difference, the chip can confirm whether foreign object exists.

**7.2.8. Other Blocks**

- Z-Tune and CLK Detector. These blocks are for bi-directional communication for power control and data exchange.
- OSC and REF. These blocks provide the timing reference and voltage reference for the whole chip.

## 8. SCHEMATIC DIAGRAM

### 8.1. Schematic diagram



**Figure 7 Typical Application Diagram**

## 8.2. Bill of Materials (BOM)

Table 9 Bill of Materials

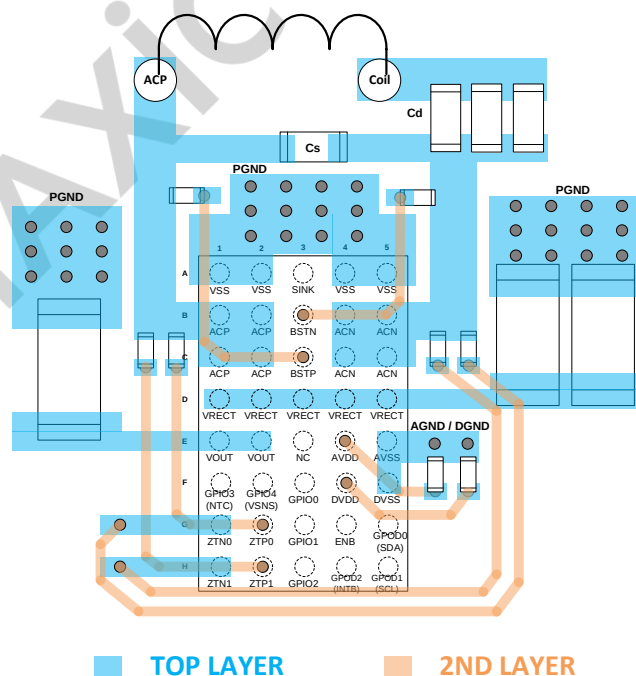
| Item | Reference       | Part                         | Footprint | Quantity |
|------|-----------------|------------------------------|-----------|----------|
| 1    | R2              | RES SMD 200R 5% 1/8W         | 0805      | 1        |
| 2    | R4,R5           | RES SMD 5.1K 1% 1/20W        | 0201      | 2        |
| 3    | R6              | RES SMD 220K 1% 1/20W        | 0201      | 1        |
| 4    | R3,R7           | RES SMD 10K 1% 1/20W         | 0201      | 1        |
| 5    | C1,C2,C3,C5,C6  | CAP CER 10 $\mu$ F 25V X7R   | 0603      | 5        |
| 6    | C8              | CAP CER 1 $\mu$ F 6.3V X7R   | 0201      | 1        |
| 7    | C9              | CAP CER 4.7 $\mu$ F 6.3V X7R | 0201      | 1        |
| 8    | C10             | CAP CER 3.3nF 50V X7R        | 0402      | 1        |
| 9    | C11,C12         | CAP CER 47nF 10V X7R         | 0402      | 2        |
| 10   | C13,C14         | CAP CER 10nF 50V X7R         | 0402      | 2        |
| 11   | C15,C16         | CAP CER 22nF 50V X7R         | 0402      | 2        |
| 12   | C17,C18,C19,C20 | CAP CER 0.1 $\mu$ F 50V X7R  | 0603      | 4        |
| 13   | C22             | CAP CER 4.7nF 50V X7R        | 0201      | 1        |
| 14   | C23             | CAP CER 22nF 25V X7R         | 0201      | 1        |
| 15   | C24             | CAP CER 680pF 10V X7R        | 0201      | 1        |
| 16   | D1              | Diode BAV21WS                | SOD-323   | 1        |
| 17   | U1              | MT5706                       | WLCSP40   | 1        |
|      |                 | Notes                        |           | 28       |

### Note

Please confirm with MAXIC during schematic design.

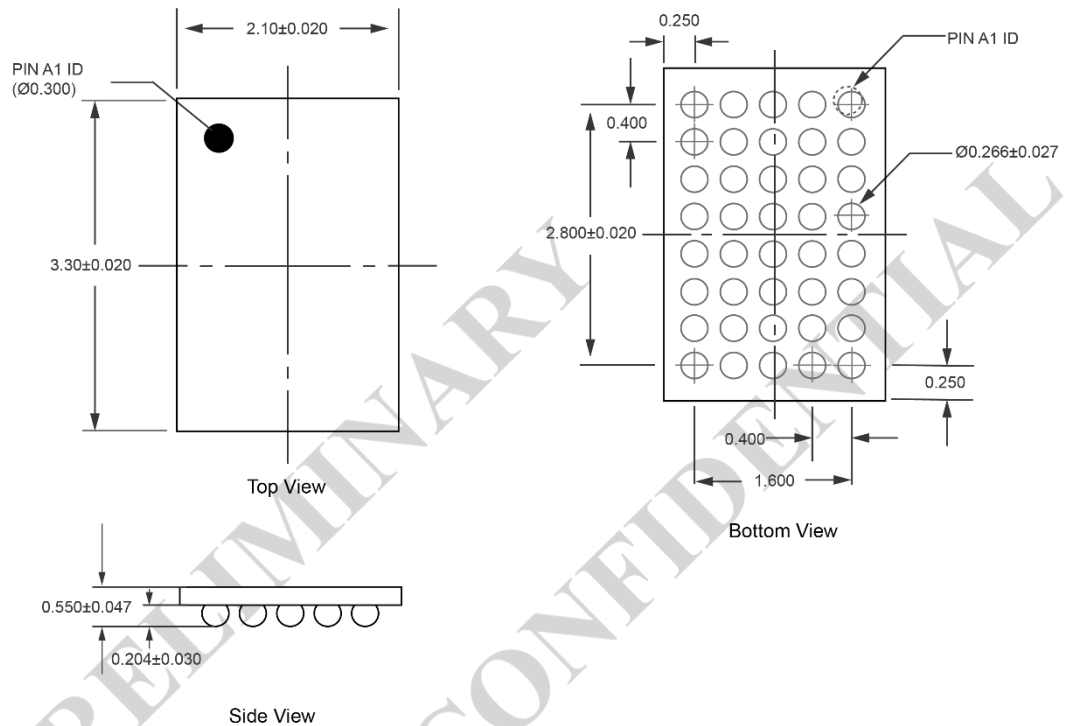
## 9. LAYOUT GUIDELINES

- The main power connections of MT5706 are VRECT, ACP, ACN, VOUT, Rx coil (LC resonant tank), and VSS (PGND). The VRECT and VOUT capacitors need to be placed as close as possible to the related pins. It is important to keep the area of the current loop, that conducts AC current from the synchronous bridge rectifier to VRECT capacitors and VSS, minimum to avoid noise, and the power traces should be as wide as possible. These connections should be routed on the same side of PCB as the chip. If need to cross layers, appropriate multiple vias are necessary.
- Heat dissipation is very important. During PCB layout, make sure that the heat on devices with severe heating can be dissipated soon. Adding internal layers improve thermal performance. Multiple vias for VSS(PGND) net are recommended to decrease the thermal resistance of the board and realize better thermal dissipation through the inner layer and power ground layers. In addition, the resonant capacitors Cd and Cs generate a lot of heat, their copper areas should be large enough to enhance heat dissipation.
- Don't connect AGND/DGND directly to the PGND on the top layer, but with more stable system ground on the internal layers through vias.
- ZTP and ZTN capacitors should be placed as close to the IC as possible. The trace from ZTN pin to the communication capacitors is best to do symmetrical with that from ZTP pin to the communication capacitor. So do BSTP and BSTN pins and their external capacitors.
- The capacitors at AVDD and DVDD pins are used to stabilize the internal linear regulators, which should be placed close to the IC.



**Figure 8 PCB layout schematic diagram**

## 10.DETAILED PACKAGING INFORMATION



### NOTES:

1. ALL DIMENSIONS AND TOLERANCES ARE PER ASME Y14.5M-1994.
2. ALL DIMENSIONS ARE IN MILLIMETERS.

Figure 9 Detailed Packaging Information

**11. ORDERING INFORMATION**

Table 10 Ordering Information

| Part No. | Package Information         | Package Quantity | Moisture Sensitivity Level |
|----------|-----------------------------|------------------|----------------------------|
| MT5706   | 3.10mm x 3.23mm<br>40-WLCSP | 3000/Tape & Reel | MSL I                      |

**12. REVISION HISTORY**

Table 11 Revision History

| Revision | Date       | Description                                               |
|----------|------------|-----------------------------------------------------------|
| 0.10     | 2022-12-30 | Preliminary version.                                      |
| 0.20     | 2023-01-05 | Add EC table, layout guideline and so on.                 |
| 0.21     | 2023-01-13 | Add detailed packaging information                        |
| 0.30     | 2023-05-19 | Upgrade EC table and ESD ratings.                         |
| 0.31     | 2023-05-23 | Modify detailed description.                              |
| 0.32     | 2023-07-04 | Upgrade EC table.                                         |
| 0.33     | 2023-07-28 | Add information of reverse charging and upgrade EC table. |
| 0.34     | 2023-09-11 | Modify detailed description.                              |
| 0.35     | 2023-10-10 | Upgrade EC table and add information of I <sup>2</sup> C. |

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